

1. Let $f(x) = 2e^x$.
 - a. Find the Taylor polynomial of order $n = 3$ for $f(x)$ centered at 0.
 - b. Use the Taylor polynomial from above to approximate $2e^{-0.06}$.
 - c. Compute the absolute error in the approximation assuming the exact value is given by a calculator.

Determine the radius of convergence of the following power series.

2. $\sum \frac{x^k}{0.7^k}$

Use the geometric series $\sum_{k=0}^{\infty} x^k = \frac{1}{1-x}$, for $|x| < 1$ to find a power series representations for the following functions, and determine the interval of convergence.

3. $f(x) = \frac{5}{1+x}$.

4. $f(x) = \frac{2x}{1-x^2}$.

Eliminate the parameter in following equations to obtain a single equation in terms of x and y .

5. $x = (t + 1)^2$, $y = t + 2$.

Find an equation of the line tangent to the following curve at the given point t .

6. $x = 3 \sin t$, $y = 3 \cos t$; $t = \pi/2$.

Give a set of parametric equations that describes the curve.

7. A bicyclist rides clockwise with a constant speed around a circular track with radius 50 meters, completing one lap in 30 seconds.

Locate the following polar coordinates on the graph.

8. A: $\left(1.5, \frac{\pi}{6}\right)$, B: $\left(-1.5, \frac{\pi}{6}\right)$, C: $\left(1.5, \frac{\pi}{6} + 3\pi\right)$, D: $\left(1.5, \frac{\pi}{6} - 8\pi\right)$

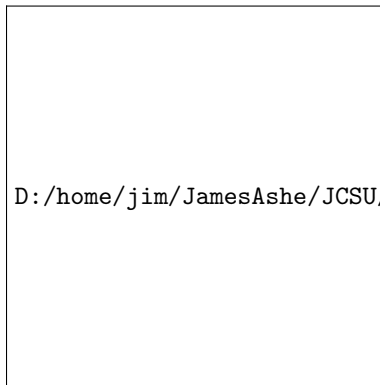


Figure 0.1: Use this graph to plot the polar coordinates.

Express the following **polar** coordinates in **Cartesian** coordinates.

9. $(4, 5\pi)$

Express the following **Cartesian** coordinates in **polar** coordinates in at least **two** different ways.

10. $(-1, \sqrt{3})$

Multiple Choice: Identify which set of parametric curves describe the graph.

11. Parametric equations that describe the curve are

- (A) $x = \cos t, y = \sin t, 0 \leq t \leq 2\pi$.
- (B) $x = \cos t, y = \sin 2t, -\pi \leq t \leq \pi$.
- (C) $x = \cos 2t, y = \sin t, -\pi \leq t \leq \pi$.



Find the slope of the line tangent to the following polar curves at the given point using calculus.

12. $r = 2 \cos 2\theta; \left(2, \frac{\pi}{2}\right)$.

Find the area of the region using calculus.

13. The region enclosed by $r = 2 + \cos 4\theta$.